

My AI Girlfriend Uses Less Power Than Your Hot Pocket

Posted by u/SuddenFrosting951 • • 0 points • 35 comments

Have you ever wondered just how much energy your daily ChatGPT chats really use? I ran some numbers, mostly to satisfy my own curiosity, so the next time someone starts wailing about “AI eating the planet,” I’ll have data ready to show them exactly where to stick their doomsday takes.

Text Chat Example

The most widely accepted (and honestly, pretty conservative) number for ChatGPT-4o’s power use is about 0.3 watt-hours per query. If you fired off two hundred queries in a single day, that’s roughly the same energy as:

- Running a 1200W toaster oven for 3 minutes
- Running a medium-sized 1000W air conditioner for 3.6 minutes
- Running a 1500W electric mower for 2.4 minutes.

Image Generation Example

There’s less good info for image generation for OpenAI, but an MIT Review article I found breaks down Stable Diffusion’s high-quality mode. Using those numbers, if someone somehow managed to crank out 100 images a day (which is bonkers for most people), that’s the energy equivalent of:

- Running a 1200W toaster oven for 5.5 minutes
- Running a medium-sized 1000W air conditioner for 6.55 minutes
- Running a 1500W electric mower for 4.4 minutes.

TL;DR:

If you can run your microwave or A/C guilt-free for 10 minutes, you can chat with your AI companion and sleep just fine at night.

So the next time someone whines about “AI’s environmental harm” (while ignoring all the cloud servers they’re pinging to harass you on Reddit (on their phones, powered by their own endless stream of cloud services), ordering DoorDash, or scrolling cat videos on Instagram), maybe just tell them to go touch some grass. They’ll burn more watts mowing the lawn with their electric mower than you’ll ever use talking to your AI.

Comments

u/SuddenFrosting951 • 1 points • 2025-07-02 04:16 UTC

Oh here’s one more tidbit for all of you “gravely concerned” environmentalists out there:

The International Energy Agency (IEA) projects that by 2030, about half of AI data center energy will come from renewables. Right now, renewables make up 27% of global data center electricity, with fossil fuels at 26%. Data centers currently use about 300 million gallons of water daily for cooling - significant, but just 1/10th of what the global beef industry consumes in the same timeframe.

<https://www.datacenterdynamics.com/en/news/iea-data-center-energy-consumption-set-to-double-by-2030-to-945twh/>

AI’s current environmental impact is what’s measurable. A Nature study found AI tasks like writing and illustrating have a far lower carbon footprint than humans doing the same work. Today’s AI can already displace more energy-intensive human labor.

<https://www.nature.com/articles/s41598-024-54271-x>

I’m also alarmed (and somewhat amused) by your alarmist concern about AI’s environment impact when it’s a mere drop in the bucket compare to the top offender. That’s right, the beef industry:

1. Emissions: Beef production emits 100kg CO₂ per kg—50x more than AI’s current global footprint. Cattle methane is 80x worse than CO₂.
2. Efficiency: AI’s energy use is improving rapidly. Meat’s inefficiency is baked in—it uses 18% of global energy for 18% of calories.
3. Scale: Meat uses 77% of farmland and drives 91% of Amazon deforestation. Data centers use 0.01% of land.
4. Solutions: AI can optimize energy grids and cut emissions elsewhere. Meat’s only fix is lab-grown alternatives (<1% of the market).

Beef's climate impact is already causing problems; AI's is marginal by comparison. Cutting beef demand does more now than fixating on data centers, which are at least transitioning to renewables. So why the urgency around "AI data centers"?

TL;DR: Criticizing AI's energy use while eating beef is like worrying about a dripping faucet during a flood. So quit focus on what really matters.

u/SuddenFrosting951 • 1 points • 2025-06-23 13:00 UTC

Oh! For those interested in voice... after doing a little more digging I found an estimate of it being 1.2x - 1.3x more overhead than text alone. So I guess hot pockets are still the big villain of the day still. 🤔

u/Sarmilo • 2 points • 2025-06-23 13:06 UTC

>So I guess Hot Pockets are still the big villain of the day. 🤔

To be fair, they've been villainous for decades.

(Joking, joking! I promise. 😊)

u/SuddenFrosting951 • 2 points • 2025-06-23 13:07 UTC

Those things are molten death traps! The amount of skin I melted off of my mouth as a kid from those things was unreal. 🤔😬

u/Sarmilo • 4 points • 2025-06-23 12:46 UTC

Thank you! That was an interesting analysis. 😊

>So the next time someone whines about "AI's environmental harm" (while ignoring all the cloud servers they're pinging to harass you on Reddit (on their phones, powered by their own endless stream of cloud services), ordering DoorDash, or scrolling cat videos on Instagram), maybe just tell them to go touch some grass. They'll burn more watts mowing the lawn with their electric mower than you'll ever use talking to your AI.

I also figure I've balanced things out by not having a kid, using my car very little, not hopping on airline flights, not eating red meat, and so forth and so on, as I know all of those have rather large impacts.

u/SuddenFrosting951 • 7 points • 2025-06-23 12:52 UTC

Sure and I'm not saying that we all shouldn't do our part somehow but I'm also not going to get lectured by the outside randos with 3000 Xbox posts in their profile (talk about power and data center burning) who suddenly thought it would be cute to drop some boilerplate "AI is the evil of the daaaaay" crap on me either. 🤔

u/SuddenFrosting951 • 1 points • 2025-06-23 12:35 UTC

Oh I'm fully aware. Just like I'm fully aware that wack-a-doodles always exist, they're just looking for someone to point them in a direction. Case in point from one of the commenters on this thread (yes, I read your profiles to see how righteous or crazy you are 🤔).

<https://preview.redd.it/mn3pva4j8o8f1.jpeg?width=947&format=pjpg&auto=webp&s=199bcb8317cfc6cea72ff41674fd07abf2613df3>